

GIFTED

MVP Technical Specification

Student-first, research-informed platform for local demo and early user testing

MVP focus: assessment quality, user experience, exploration missions, Gifted Passport, and parent value

Parameter	MVP Decision
Primary users	Students aged 12-17
Secondary users	Parents
Internal users	Admin / Content Manager
Frontend	React + TypeScript + Vite
Backend	Django + Django REST Framework
Database	PostgreSQL
Media	Local files for MVP
Architecture	Modular Monolith
Deployment	Local demo / controlled testing

Version 1.0

Executive Summary

MVPning vazifasi productionga tayyor katta platforma qurish emas. Asosiy maqsad - real foydalanuvchilar Giftedning qiymatini ishlatib ko'rishi, assessment va exploration flow ishlashini tekshirish, Gifted Passportni tushunarli qilish va parent segmentidagi qiziqishni o'lchash.

Assessment discovers -> Mission validates -> Opportunity exposes -> Evidence updates -> Passport evolves -> Recommendation improves

1. MVPda nimani tekshiramiz?

Hypothesis	Tekshiriladigan narsa
Young people assessmentni oxirigacha ishlaydimi?	Completion
Interactive tasks oddiy quizdan qiziqroqmi?	Engagement
Gifted Passport tushunarlimi?	Comprehension
Recommendations meaningful ko'rinadimi?	Relevance
Parents bunday productga qiziqish bildiradimi?	Parent engagement

2. MVP Architecture

MVP uchun React frontend + Django/DRF backend + PostgreSQL + local media yetadi.

```
GIFTED MVP
|
React + TypeScript + Vite
|
REST API
|
Django + DRF - Modular Monolith
|
PostgreSQL

Media: backend/media/
Static: frontend/public/
```

- Hozircha S3 yo'q
- Redis yo'q
- Celery yo'q
- Microservices yo'q
- Elasticsearch yoki alohida analytics DB yo'q

Arxitektura keyinchalik production komponentlarni qo'shishga imkon beradigan qilib tartibli quriladi.

3. Backend structure

```
backend/
├── config/
├── apps/
│   ├── accounts/
│   ├── learners/
│   ├── parents/
│   ├── assessments/
│   ├── questions/
│   ├── signals/
│   ├── evidence/
│   ├── passports/
│   ├── missions/
│   ├── opportunities/
│   └── recommendations/
```

```

├── analytics/
├── administration/
├── media/
├── static/
├── manage.py
└── requirements.txt

```

Bu modular monolith: har business domain alohida modul, lekin bitta Django project ichida.

4. Frontend structure

```

frontend/
├── src/
│   ├── app/
│   ├── api/
│   ├── components/
│   ├── layouts/
│   ├── features/
│   │   ├── auth/
│   │   ├── assessment/
│   │   ├── passport/
│   │   ├── missions/
│   │   ├── opportunities/
│   │   └── parent/
│   ├── pages/
│   │   ├── student/
│   │   ├── parent/
│   │   └── admin/
│   ├── hooks/
│   ├── types/
│   └── utils/

```

Student, parent va admin UI bitta frontend ichida bo'ladi, route va permissionlar role bo'yicha ajratiladi.

5. User roles

Role	Purpose
STUDENT	Main Gifted experience
PARENT	Child progress and recommendations
ADMIN	Content, users, basic analytics

Keyinchalik MENTOR, SCHOOL_MANAGER va LEARNING_CENTER_MANAGER rollari qo'shilishi mumkin.

6. Main Student Journey

```

Landing
↓
Register / Login
↓
Basic Profile
↓
Onboarding
↓
Discovery Assessment
↓
Initial Gifted Passport
↓
Recommended Exploration
↓
Mission / Case / Mini Lab
↓
New Evidence

```

↓
Updated Passport
↓
Recommended Opportunities

7. Student Dashboard

Oldin ishlab chiqilgan Gifted Personal Passport dashboard student panelning asosiy UX reference'i bo'ladi.

Overview
My Passport
Assessments
Explore
Opportunities
Growth
Goals
Profile

8. Student Dashboard - Overview

- Hero: user photo, name, age, city, profile completion, short Gifted message
- Overview cards: Interests, Reasoning, Exposure, Evidence
- Primary CTA: Continue your discovery yoki Try your next mission

9. Onboarding

- Step 1: Name, Age, Gender (optional), City, Language
- Step 2: I know what I want / I have several ideas / I'm not sure / I have no idea yet
- Step 3: Which activities have you already tried? - visual quick choices

Onboarding maksimal 3-4 minut bo'lishi kerak.

10. Assessment Engine

MVPning yuragi: Interests, Aptitude/Reasoning, Work Style, Values, Exposure, Real-world Evidence.

Bu signallar bitta umumiy scorega majburan birlashtirilmaydi.

11. Assessment Session Structure

1. Interests
2. Visual choices
3. Reasoning games
4. Situational cases
5. Values
6. Exposure

Har section taxminan 3-6 minut. Full assessment 20-30 minut. Save & Continue Later qo'llab-quvvatlanadi.

12. Interests Module

MVPda 24-30 item. RIASECga mapped: Realistic, Investigative, Artistic, Social, Enterprising, Conventional.

- Activity choice
- Picture pair
- Would you try this?
- Forced choice

13. Picture Choice Module

Userga ikki rasm ko'rsatiladi va qaysi faoliyatni sinab ko'rishni xohlashi so'raladi.

Question.type = IMAGE_PAIR. Media local /media/questions/ ichida saqlanadi. Admin image upload qila oladi.

14. Aptitude / Reasoning Module

MVPda 4-5 dimension: Pattern reasoning, Spatial reasoning, Logical reasoning, Numerical reasoning, Sequential reasoning.

Har biridan 4-6 task, jami taxminan 20-25 task.

15. Interactive task types

```
PatternTask
ImageChoiceTask
SequenceTask
SpatialChoiceTask
MultipleChoiceTask
SituationalCaseTask
SliderChoiceTask
```

Frontend Question.type bo'yicha tegishli componentni render qiladi. Har savol uchun alohida hardcoded page qilinmaydi.

16. Work Style - Situational Cases

MVPda 10-12 case. Bir javob bitta labelga teng emas; har option bir nechta signalga weighted mapping qilishi mumkin.

17. Values

MVPda 10-12 forced-choice item.

Asosiy dimensions: Impact, Autonomy, Learning, Achievement, Creativity, Security, Recognition, Collaboration.

18. Exposure Map

User Coding, Design, Science experiments, Business, Teaching, Debate, Photography, Robotics, Writing, Volunteering, Public speaking, Music, Engineering kabi activitylarni qanchalik ko'rgan yoki sinaganini belgilaydi.

```
Never seen
Know about it
Tried once
Tried several times
Do regularly
```

19. Initial Gifted Passport

Assessment tugashi bilan initial Passport hosil bo'ladi.

- Interests - RIASEC pattern
- Reasoning signals - current observed strengths
- Work Style - exploratory tendencies
- Exposure - High / Medium / Low
- Real-world Evidence - key activities

20. Confidence

Har signal uchun Low / Medium / High confidence ko'rsatilishi mumkin.

Misol: Product Design Interest - Medium confidence. More exploration needed.

21. Exploration Engine

Assessmentdan keyin usarga 3 ta recommended exploration ko'rsatiladi.

Har recommendationda Why? bo'limi bo'ladi: qaysi signal va exposure gap asos bo'lganini tushuntiradi.

22. Missions

MVP uchun 6-8 high-quality mission yetadi.

- Design a better school bag - creativity, empathy, prioritization, design interest
- Save the event budget - numerical reasoning, decision making, business exposure
- Investigate the science mystery - scientific reasoning, curiosity, persistence

23. Mission flow

```

Mission Intro
  ↓
Context
  ↓
Task
  ↓
Decision
  ↓
Feedback
  ↓
Reflection
  ↓
Evidence Created

```

24. Mini Virtual Labs

- Engineering Lab - bridge challenge: material, structure, cost
- Product Design Lab - user problem, feature selection, prototype decision
- Business Lab - budget allocation, channel choices, result

Bu full simulation engine emas; React interactive components ko'rinishida bo'ladi.

25. Evidence

Har mission tugashi yangi Evidence yaratadi. Masalan: Completed Design Mission -> Problem Solving, Creativity, User Empathy signals.

26. Evidence model

```

Evidence
- id
- learner
- source_type
- source_id
- title
- description
- created_at

SignalEvidence
- evidence
- signal
- value

```

- confidence

27. Passport update

MVPda background queue shart emas. Mission tugaganda update_passport(user) service chaqiriladi.

Business logic view ichiga yozilmaydi; services/passport_service.py kabi service layerda bo'ladi.

28. Recommendations

MVPda black-box AI emas, tushunarli weighted rules ishlatiladi.

Product Design	
Artistic interest	+3
Problem solving	+2
Visual reasoning	+2
Creativity evidence	+3
Design exposure low	+1 exploration priority

29. Recommendation 'Why?'

Majburiy feature. User recommendation nima uchun chiqqanini ko'radi.

- Strong creative interest signal
- Good visual reasoning
- Enjoyed problem-solving tasks
- Real-world exposure is still low - try a design mission

30. Opportunities

MVPda real content: Courses, Events, Competitions, Programs, Scholarships.

20-40 ta seed opportunity demo va user testing uchun yetadi.

31. Opportunity page

- Title
- Type
- Organization
- Age
- Format
- Deadline
- Location

Detail page: Description, Who is it for?, What will you explore?, Application link.

32. Opportunity recommendations

Passport signal va exposure gap asosida Recommended for you blocki chiqadi.

Misol: Robotics Competition - matches Problem Solving + Investigative Interest; useful because engineering exposure is low.

33. Save opportunity

Student opportunityni save qila oladi. Parent ham saved opportunitylarni ko'rishi mumkin.

34. Parent Panel

Parent MVPdagi ikkinchi asosiy user. Marketing va monetizatsiya nuqtai nazaridan parent buyer/customer bo'lishi mumkin.

35. Parent onboarding

Parent register qiladi va childga connection code orqali ulanadi. Misol: GFT-48291.

36. Parent Dashboard

My Child
Overview
Potential Map
Growth
Evidence
Opportunities
Recommendations

37. Parent Home

- Hero: child name, age, Passport completion
- What we're learning: emerging interests, strong signals, low exposure areas
- Recent growth: completed mission, tried lab, saved event
- Next opportunities: recommended activities

38. Parent UX principles

Parentga keskin label berilmaydi. 'Your child is bad at math' yoki 'definitely artistic' kabi xulosalar chiqarilmaydi.

Tili: We are currently seeing..., There is not enough evidence yet..., Trying these experiences may help clarify...

39. Parent Insight Card

Misol: Hidden potential - learner engineeringga juda kam exposure olgan, shuning uchun low current interest hali low potential deb talqin qilinmaydi.

CTA: Explore engineering activities.

40. Parent Recommendations

Parentga Try next blokida workshop, mission yoki opportunity tavsiya qilinadi va why explanation ko'rsatiladi.

41. Parent weekly summary - MVP optional

Notification tizimisiz dashboard ichida: 2 activities completed, 1 new evidence added, 1 opportunity saved kabi summary.

42. Parent permissions

Parent can see	Default restricted
Passport summary	Individual raw answers
Growth	Private reflection text
Opportunities	
Recommendations	
Evidence summary	

43. Admin Panel - MVP scope

Admin ikkinchi darajali. Django Admin + oddiy custom admin page yetadi; alohida katta analytics product qurilmaydi.

44. Admin capabilities

- Users: view students, parents, deactivate account
- Questions: create, edit, delete, publish
- Missions: create, edit, publish
- Opportunities: create Course/Event/Competition/Program/Scholarship
- Basic Analytics: users, assessment starts/completions, mission completions

45. Opportunity Admin

```
Opportunity
- title
- type
- description
- organization
- age_min / age_max
- online / city
- start_date / deadline
- external_url
- image
- status
```

```
TYPE:
COURSE
EVENT
COMPETITION
PROGRAM
SCHOLARSHIP
```

Images MVPda /media/opportunities/ ichida local saqlanadi.

46. Admin content form

Title, Description, Type, Age, Category, Signals, Deadline, URL, Image, Publish.

47. Assessment Content Admin

```
Question
- type
- construct
- text
- difficulty
- image
- status
- order
```

```
QuestionOption
- question
- text
- image
- value
```

48. Research source fields

Har content item uchun research_note, source_name, source_url kabi fieldlar bo'lishi mumkin. Bu savollar random bo'lib ketmasligi va content provenance saqlanishi uchun.

49. Core Database - User

```
User
- id
- email
- password
- role
- is_active
```

50. LearnerProfile

```
LearnerProfile
- user
- name
- age
- city
- avatar
- bio
```

51. ParentProfile

```
ParentProfile
- user
- name
```

52. ParentChild

```
ParentChild
- parent
- child
- status
```

53. Assessment models

```
Assessment
AssessmentSection
Question
QuestionOption
AssessmentSession
AssessmentResponse
```

54. Signals

```
Signal
- name
- category
```

CATEGORY:
 INTEREST
 APTITUDE
 WORK_STYLE
 VALUE
 EXPOSURE

ResponseSignalMap
 - question_option
 - signal
 - weight

55. Passport models

Passport
 - learner
 - version
 - updated_at

PassportSignal
 - passport
 - signal
 - score
 - confidence
 - evidence_count

56. Mission models

Mission
 MissionStep
 MissionAttempt
 MissionResponse

57. Opportunity models

Opportunity
 OpportunitySignal
 SavedOpportunity

58. Recommendation model

Recommendation
 - learner
 - type
 - object_id
 - score
 - reason
 - created_at

TYPE:
 MISSION
 OPPORTUNITY
 DOMAIN

59. Basic analytics

```
AnalyticsEvent
- user
- event_name
- object_type
- object_id
- created_at

EVENTS:
login
assessment_started
assessment_completed
mission_started
mission_completed
opportunity_viewed
opportunity_saved
parent_connected
```

60. API base

Base path: /api/v1/

61. Auth API

```
POST /auth/register/
POST /auth/login/
POST /auth/logout/
GET /auth/me/
```

62. Student API

```
GET /student/dashboard/
GET /student/profile/
PATCH /student/profile/
```

63. Assessments API

```
GET /assessments/
POST /assessments/{id}/start/
GET /assessment-sessions/{id}/
POST /assessment-sessions/{id}/answer/
POST /assessment-sessions/{id}/complete/
```

64. Passport API

```
GET /passport/
GET /passport/signals/
GET /passport/evidence/
GET /passport/growth/
```

65. Missions API

```
GET    /missions/
GET    /missions/recommended/
GET    /missions/{id}/
POST   /missions/{id}/start/
POST   /mission-attempts/{id}/answer/
POST   /mission-attempts/{id}/complete/
```

66. Opportunities API

```
GET    /opportunities/
GET    /opportunities/{id}/
GET    /opportunities/recommended/
POST   /opportunities/{id}/save/
DELETE /opportunities/{id}/save/
```

67. Parent API

```
POST /parent/connect/
GET  /parent/children/
GET  /parent/children/{id}/overview/
GET  /parent/children/{id}/passport/
GET  /parent/children/{id}/growth/
GET  /parent/children/{id}/recommendations/
```

68. Admin API

```
GET/POST /admin-api/opportunities/
GET/POST /admin-api/missions/
GET/POST /admin-api/questions/
GET      /admin-api/analytics/
```

69. Student frontend routes

```
/
/login
/register
/app
/app/passport
/app/assessment
/app/assessment/:sessionId
/app/explore
/app/missions/:id
/app/opportunities
/app/opportunities/:id
/app/growth
/app/goals
/app/profile
```

70. Parent routes

```
/parent
```

```

/parent/connect
/parent/child/:id
/parent/child/:id/passport
/parent/child/:id/growth
/parent/child/:id/opportunities

```

71. Admin routes

```

/admin
/admin/users
/admin/questions
/admin/missions
/admin/opportunities
/admin/analytics

```

72. Performance - sayt qotmasligi uchun

MVPda S3/Redis shart emas, lekin database query discipline majburiy.

- select_related()
- prefetch_related()
- Pagination: default 20 items/page

73. Dashboard API

Frontend 15 endpointni parallel urmasin. Bitta /student/dashboard/ endpoint summary qaytaradi.

```

{
  "profile_completion": 62,
  "top_signals": [],
  "next_mission": {},
  "recent_evidence": [],
  "recommended_opportunities": []
}

```

74. Detail lazy-loading

Dashboardda summary, detail faqat user View all yoki detail page ochganda yuklanadi.

75. Images

Local media yetadi, lekin upload size limit, image resize va WebP tavsiya qilinadi. Masalan max upload = 2MB.

76. Frontend performance

Student, Parent va Admin route chunklari lazy-loaded bo'ladi. Heavy virtual lab component faqat lab ochilganda yuklanadi.

77. State management

MVPda Redux majburiy emas. Server state uchun TanStack Query, local UI state uchun useState / Context yetadi.

78. Authentication

MVPda JWT: access token + refresh token. Password hashing Django orqali.

79. Local setup

```
Backend:
python manage.py runserver

Frontend:
npm run dev

Database:
Local PostgreSQL
```

80. Demo seed command

Bitta seed command demo student, demo parent, assessment questions, missions va opportunities yaratadi.

```
python manage.py seed_demo
```

81. Demo accounts

```
student@gifted.demo
password: demo123

parent@gifted.demo
password: demo123

admin@gifted.demo
password: demo123
```

Faqat local demo uchun.

82. MVP content target

Content	Target
Interest items	24-30
Picture choices	12-18
Reasoning tasks	20-25
Situational cases	10
Values choices	10
Exposure activities	20-30
Missions	6
Virtual mini-labs	3
Opportunities	20-40

83. MVP priority - P0 MUST HAVE

Area	Must have
Student	Auth, Onboarding, Assessment, Passport, Recommendations, Missions, Opportunities, Growth
Parent	Connect child, Overview, Passport summary, Growth, Recommendations, Opportunities
Admin	Questions, Missions, Opportunities, Users, Basic analytics

84. P1 - if time remains

- Goals
- Opportunity favorites

- Assessment resume
- Mission history
- Parent weekly summary
- Better analytics

85. P2 - post-MVP

- Mentors
- School dashboards
- Learning center dashboards
- AI coach / Chat
- Advanced recommendation engine
- University matching
- Mobile application
- Notifications / Email / Telegram
- S3 / Celery / Redis

86. Eng muhim product loop

```

DISCOVER
Assessment
↓
EXPLORE
Mission / Event / Course
↓
VALIDATE
Real Evidence
↓
DEVELOP
New experiences and skills
↓
GUIDE
Better recommendations
↓
PASSPORT UPDATE
↻

```

87. Student uchun birinchi demo flow

1. Student login qiladi va Gifted Dashboard ochiladi.
2. Assessmentni boshlaydi: text + pictures + reasoning.
3. Assessment tugaydi va Initial Gifted Passport hosil bo'ladi.
4. System kuchli signals va low exposure bo'lgan yo'nalishni ko'rsatadi.
5. Student mini Product Design missionini bajaradi.
6. Mission tugagach Evidence added bo'ladi.
7. Passport update bo'ladi va confidence o'zgaradi.
8. Recommended Opportunity ko'rsatiladi.
9. Parent login qilib progressni ko'radi.

Bu 7-10 minutlik demo Giftedning asosiy product loopini to'liq tushuntirishi kerak.

88. Parent uchun marketing hook

Parent panel faqat scorelarni ko'rsatmaydi. Asosiy message: "Don't choose a course before understanding what your child has actually had the chance to discover."

Parent panel: Potential signals + Exposure gaps + What to try next.

89. Definition of Done - Student

- Register/login qiladi
- Assessmentni tugatadi
- Initial Passport hosil bo'ladi
- Recommendation ko'radi
- Mission bajaradi
- Evidence qo'shiladi
- Passport o'zgaradi
- Opportunity ko'radi va save qiladi

90. Definition of Done - Parent

- Register qiladi
- Child bilan ulanadi
- Passport summary ko'radi
- Growth ko'radi
- Recommendation ko'radi
- Opportunity ko'radi

91. Definition of Done - Admin

- Assessment content qo'shadi
- Mission qo'shadi
- Course / event / competition qo'shadi
- Userlarni ko'radi
- Basic analytics ko'radi

92. Final MVP Architecture

```

GIFTED MVP
|
+-- STUDENT
+-- PARENT
+-- ADMIN
|
React + TypeScript
|
Django API
|
PostgreSQL
/      \
Local Media Seed Data

```

93. Final Product Logic

```

Assessment
↓
Signals
↓
Passport
↓
Exploration
↓
Mission / Opportunity
↓
Evidence
↓

```

Passport Update



Better Recommendation

Bu versiya MVP uchun yetarlicha professional, lekin ortiqcha murakkab emas. Resursning katta qismi infrastructure complexityga emas, assessment sifati, cases, visuals, missions, Passport UX va parent insightga sarflanadi.

GIFTED MVP

Discover -> Explore -> Validate -> Develop -> Guide

MVPda infrastruktura murakkabligiga emas, assessment sifati, real exploration, Gifted Passport, parent insight va product validationga ustuvorlik beriladi.